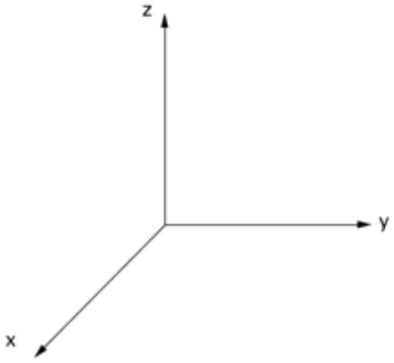
Equation of a Line in R^3

Similar to the derivation in R^2



Vector Equation in R^3

Parametric Equations in R^3

Symmetric Equations in R^3

Note: A line in R^3 does not have a single unique _____.
Therefore, a line in R^3 does not have a _____ equation.

Example 1: Determine the vector, parametric and symmetric equations of a line passing through $P(1,-4,3)$ and $Q(2,-1,-5)$.

Example 2: Classify the following two lines as equivalent, parallel, perpendicular or none of the above.

$\vec{r}_1 = (1,-4,3) + t(1,3,-8)$, $t \in \Re$

$\vec{r}_2 = (2,-1,-5) + s(-2,-6,16)$, $s \in \Re$

Example 3: Determine the symmetric equations of a line that goes through the point $(3,-1,2)$ with the direction vector $\vec{m} = (4,0,-5)$.